

# PROS1-A480-OSB

Serie PROS · R1234ZE · senza serbatoio / without tank



PROS1-A480-OSB

RESA FRIGORIFERA · COOLING CAPACITY

312 kW

EER · 12/7 °C · AIR 35 °C

2,92

RESA NOMINALE · 15 °C · AIR 25 °C

475 kW

PORTATA ACQUA · WATER FLOW

53,7 m³/h

|                            |                 |                                        |                       |
|----------------------------|-----------------|----------------------------------------|-----------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 4.800 × 2.197 × 2.402 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                     |
| Ventilatori / Fans         | 8               | Taglia / Size                          | 480                   |
| Refrigerante / Refrigerant | R1234ZE · GWP 7 | Campo aria esterna / Ambient air range | -5 / +50 °C           |

## Resa in sintesi / Capacity at a glance kW

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C | 50 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 348   | 331   | 312   | 294   | 275   | 255   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 475   | 452   | 428   | 404   | 379   | 354   |

Refrigeratori centralizzati condensati ad aria con compressori a vite, per le grandi potenze industriali e commerciali, con parzializzazione della potenza 25-50-75-100 %.

Centralised air-cooled chillers with screw compressors for large industrial and commercial loads, with 25-50-75-100 % capacity steps.

# Dati tecnici

## Technical data

PROS1-A480-0SB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                         |
|-------------------------------------|-------------------------|
| Resa frigorifera / Cooling capacity | 474,84 kW               |
| Resa frigorifera / Cooling capacity | 408.362 kcal/h          |
| Potenza assorbita / Power input     | 91,10 kW                |
| EER / EER                           | 5,21                    |
| Portata acqua / Water flow          | 81,67 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                         |
|-------------------------------------|-------------------------|
| Resa frigorifera / Cooling capacity | 312,38 kW               |
| Potenza assorbita / Power input     | 107,03 kW               |
| EER / EER                           | 2,92                    |
| ESEER / ESEER                       | 3,94                    |
| Portata acqua / Water flow          | 53,73 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                       |
|---------------------------------------------|-----------------------|
| Compressori / Compressors                   | 2 × Vite              |
| Circuiti frigoriferi / Refrigerant circuits | 2                     |
| Ventilatori / Fans                          | 8                     |
| Condensatore / Condenser                    | a microcanali         |
| Evaporatore / Evaporator                    | a fascio tubiero      |
| Refrigerante / Refrigerant                  | R1234ZE · GWP 7 · A2L |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 4.800 × 2.197 × 2.402 mm  |
| Campo aria esterna / Ambient air range                                                    | -5 / +50 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 · 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 306,1 kW     |
| MEG 30 % | × 0,950 | 296,8 kW     |
| MEG 40 % | × 0,920 | 287,4 kW     |
| MPG 20 % | × 0,975 | 304,6 kW     |
| MPG 30 % | × 0,960 | 299,9 kW     |
| MPG 40 % | × 0,950 | 296,8 kW     |

# Prestazioni Performance

PROS1-A480-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35               | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|------------------|-----|-----|-----|
| 1             | 299 | 286 | 272 | 257 | 243              | 228 | 212 | 197 |
| 2             | 312 | 298 | 284 | 269 | 254              | 238 | 222 | 206 |
| 3             | 325 | 310 | 296 | 280 | 264              | 248 | 232 | 215 |
| 4             | 339 | 324 | 308 | 292 | 276              | 259 | 242 | 225 |
| 5             | 352 | 337 | 321 | 305 | 288              | 270 | 252 | 234 |
| 6             | 367 | 351 | 335 | 318 | 300              | 282 | 264 | 245 |
| 7             | 382 | 365 | 348 | 331 | <span>312</span> | 294 | 275 | 255 |
| 8             | 397 | 380 | 363 | 344 | 326              | 306 | 287 | 267 |
| 9             | 413 | 395 | 377 | 358 | 339              | 319 | 298 | 278 |
| 10            | 430 | 411 | 392 | 373 | 353              | 332 | 311 | 290 |
| 11            | 446 | 427 | 408 | 388 | 367              | 345 | 324 | 302 |
| 12            | 464 | 444 | 424 | 403 | 382              | 360 | 337 | 314 |
| 13            | 481 | 461 | 440 | 419 | 397              | 374 | 351 | 327 |
| 14            | 500 | 479 | 458 | 435 | 412              | 389 | 365 | 340 |
| 15            | 519 | 497 | 475 | 452 | 428              | 404 | 379 | 354 |
| 16            | 538 | 516 | 493 | 469 | 445              | 420 | 394 | 368 |
| 17            | 558 | 535 | 511 | 487 | 462              | 436 | 409 | 382 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15 | 20 | 25 | 30  | 35               | 40  | 45  | 50  |
|---------------|----|----|----|-----|------------------|-----|-----|-----|
| 1             | 73 | 80 | 87 | 96  | 105              | 115 | 127 | 139 |
| 2             | 73 | 80 | 88 | 96  | 106              | 116 | 127 | 140 |
| 3             | 73 | 80 | 88 | 96  | 106              | 116 | 128 | 140 |
| 4             | 74 | 81 | 88 | 97  | 106              | 117 | 128 | 141 |
| 5             | 74 | 81 | 88 | 97  | 106              | 117 | 128 | 141 |
| 6             | 74 | 81 | 89 | 97  | 107              | 117 | 129 | 142 |
| 7             | 74 | 81 | 89 | 98  | <span>107</span> | 118 | 129 | 142 |
| 8             | 75 | 82 | 89 | 98  | 107              | 118 | 129 | 142 |
| 9             | 75 | 82 | 89 | 98  | 108              | 118 | 130 | 143 |
| 10            | 75 | 82 | 90 | 98  | 108              | 118 | 130 | 143 |
| 11            | 75 | 82 | 90 | 99  | 108              | 119 | 130 | 143 |
| 12            | 76 | 83 | 90 | 99  | 108              | 119 | 131 | 144 |
| 13            | 76 | 83 | 91 | 99  | 109              | 119 | 131 | 144 |
| 14            | 76 | 83 | 91 | 99  | 109              | 120 | 131 | 144 |
| 15            | 77 | 83 | 91 | 100 | 109              | 120 | 132 | 145 |
| 16            | 77 | 84 | 91 | 100 | 109              | 120 | 132 | 145 |
| 17            | 77 | 84 | 92 | 100 | 110              | 120 | 132 | 145 |

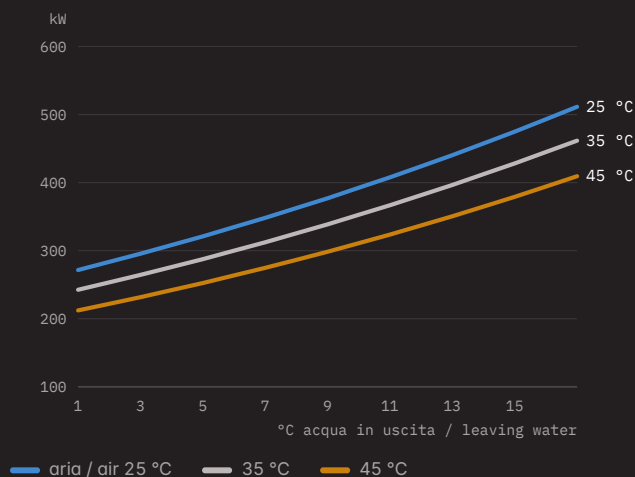
## EER / EER (kW/kW)

| LWT ↓ · air → | 15                | 20   | 25   | 30   | 35                | 40   | 45   | 50   |
|---------------|-------------------|------|------|------|-------------------|------|------|------|
| 1             | <span>4,10</span> | 3,58 | 3,11 | 2,68 | 2,31              | 1,97 | 1,67 | 1,41 |
| 2             | <span>4,27</span> | 3,72 | 3,23 | 2,80 | 2,40              | 2,05 | 1,74 | 1,47 |
| 3             | <span>4,43</span> | 3,87 | 3,36 | 2,91 | 2,50              | 2,14 | 1,81 | 1,53 |
| 4             | <span>4,60</span> | 4,02 | 3,50 | 3,02 | 2,60              | 2,22 | 1,89 | 1,60 |
| 5             | <span>4,77</span> | 4,17 | 3,63 | 3,14 | 2,70              | 2,31 | 1,97 | 1,66 |
| 6             | <span>4,95</span> | 4,33 | 3,77 | 3,27 | 2,81              | 2,41 | 2,05 | 1,73 |
| 7             | <span>5,14</span> | 4,49 | 3,91 | 3,39 | <span>2,92</span> | 2,50 | 2,13 | 1,80 |
| 8             | <span>5,33</span> | 4,66 | 4,06 | 3,52 | 3,03              | 2,60 | 2,21 | 1,87 |
| 9             | <span>5,51</span> | 4,83 | 4,21 | 3,65 | 3,15              | 2,70 | 2,30 | 1,95 |
| 10            | <span>5,71</span> | 5,01 | 4,37 | 3,79 | 3,27              | 2,80 | 2,39 | 2,02 |
| 11            | <span>5,91</span> | 5,19 | 4,53 | 3,93 | 3,39              | 2,91 | 2,48 | 2,10 |
| 12            | <span>6,12</span> | 5,38 | 4,70 | 4,08 | 3,52              | 3,02 | 2,58 | 2,19 |
| 13            | <span>6,33</span> | 5,57 | 4,86 | 4,23 | 3,65              | 3,13 | 2,67 | 2,27 |
| 14            | <span>6,55</span> | 5,76 | 5,04 | 4,38 | 3,79              | 3,25 | 2,78 | 2,36 |
| 15            | <span>6,76</span> | 5,96 | 5,21 | 4,54 | 3,92              | 3,37 | 2,88 | 2,44 |
| 16            | <span>6,99</span> | 6,16 | 5,40 | 4,70 | 4,07              | 3,50 | 2,99 | 2,54 |
| 17            | <span>7,21</span> | 6,36 | 5,58 | 4,86 | 4,21              | 3,62 | 3,10 | 2,63 |

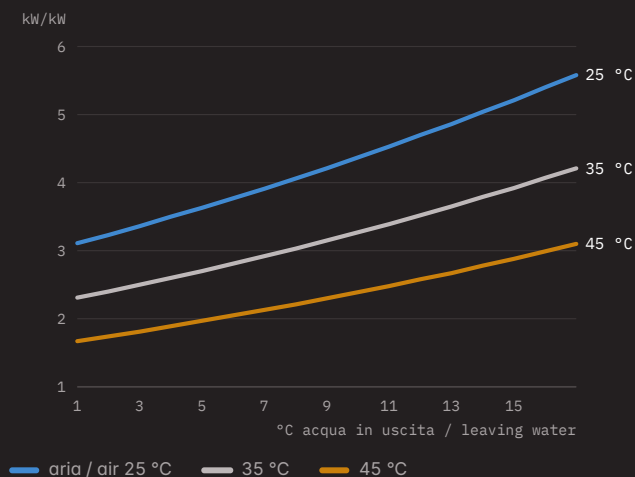
# Curve e dimensioni

## Curves and dimensions

### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 4.800 × 2.197 × 2.402 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/3 Options

PROS1-A480-OSB · 75 optional disponibili per questa configurazione · 75 options available for this configuration

| Pompe / Pumps                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pompa 71 m³/h – prevalenza 25 m<br>Pump 71 m³/h – head 25 m                                                                                                           |
| Pompa 71 m³/h – prevalenza 40 m<br>Pump 71 m³/h – head 40 m                                                                                                           |
| Pompa doppia montata 71 m³/h – prevalenza 25 m<br>Twin pump, factory-fitted, 71 m³/h – head 25 m                                                                      |
| Pompa doppia montata 71 m³/h – prevalenza 40 m<br>Twin pump, factory-fitted, 71 m³/h – head 40 m                                                                      |
| Serbatoi e vasi di espansione / Tanks and expansion vessels                                                                                                           |
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 1000 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 1500 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa doppia montata<br>External water tank 1000 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa doppia montata<br>External water tank 1500 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia montata<br>External water tank 2000 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory-fitted twin pump                         |
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 1000 l, AISI 304 with factory-fitted single AISI 304 pump |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 1500 l, AISI 304 with factory-fitted single AISI 304 pump |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory-fitted single AISI 304 pump |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory-fitted single AISI 304 pump |

|                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory-fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa singola montata<br>External water tank 1000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola montata<br>External water tank 1500 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata<br>External water tank 2000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory-fitted single pump                                           |
| Serbatoio idraulico esterno 1000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 1000 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 1500 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 2000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 2000 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 3000 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 5000 l, pressurised, galvanised steel with factory-fitted twin pump    |
| Serbatoio idraulico esterno 1000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 1000 l, pressurised, galvanised steel with factory-fitted single pump |
| Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 1500 l, pressurised, galvanised steel with factory-fitted single pump |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory-fitted single pump |

# Optional — 2/3 Options

PROS1-A480-OSB · 75 optional disponibili per questa configurazione · 75 options available for this configuration

## Serbatoi e vasi di espansione / Tanks and expansion vessels

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump

Vaso di espansione 100 l

Expansion vessel 100 l

Vaso di espansione 200 l

Expansion vessel 200 l

Vaso di espansione 300 l

Expansion vessel 300 l

Vaso di espansione 500 l

Expansion vessel 500 l

Vaso di espansione 750 l

Expansion vessel 750 l

## Valvole e filtri / Valves and filters

Elettrovalvola 2"

Solenoid valve 2"

Elettrovalvola 3"

Solenoid valve 3"

Elettrovalvola 4"

Solenoid valve 4"

Elettrovalvola 5"

Solenoid valve 5"

Elettrovalvola 6"

Solenoid valve 6"

Filtro a dischi in acciaio 2"

Steel disc filter 2"

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y-strainer 2"

Filtro meccanico a Y 3"

Y-strainer 3"

Filtro meccanico a Y 4"

Y-strainer 4"

Filtro meccanico a Y 5"

Y-strainer 5"

Filtro meccanico a Y 6"

Y-strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

## Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

## Free cooling / Free cooling

Free cooling 400 kW – 8 ventilatori (rif. 6)

Free cooling 400 kW – 8 fans (ref. 6)

## Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

## Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

## Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

# Optional — 3/3

## Options

PROS1-A480-OSB · 75 optional disponibili per questa configurazione · 75 options available for this configuration

### Quadro elettrico / Electrical panel

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
- Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
- Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
- Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
- Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
- With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
- Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
- Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
- Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

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Continuity by design